Quadratics, gently

Summary

This section summarizes what you want to remember after you're finished reading. Sometimes called TL;DR (too long didn't read). Skim or skip this part if you are new to the topic.

The general equation for a quadratic is

$$y = ax^2 + bx + c$$

where a, b and c are constants.

◦ a determines how steeply the sides of parabola curve and its sign tells in what direction the "cup opens", up or down.

◦ c shifts the parabola's vertical position and is also the y-value when $x = 0$.

◦ b and a together determine where the pointy end lies.

This point is called the vertex of the parabola and it is always the minimum for y (or maximum if the parabola opens down). We will call its x-value m for minimum, and just give the result, for now.

$$m = -\frac{b}{2a}$$

To "solve" a quadratic means to find its zeros, or roots. These are the

x-values that make the whole thing equal to zero.

$$ax^2 + bx + c = 0$$

Dividing by a does not change the roots. The same values of x satisfy

$$x^2 + \frac{b}{a}x + \frac{c}{a} = 0$$

The zeros can always be found using the quadratic formula. Here is a simple version of that:

$$x = m \pm \sqrt{m^2 - \frac{c}{a}}$$

You must memorize both the formula for m and this one.

Sometimes we can guess the zeros since

$$(x - s)(x - t) = x^2 - (s + t)x + st = 0$$

See if you can two factors that add to give negative b and multiply to give c.

As an example

$$x^2 + 2x - 63 = y$$

Find integers two units apart that multiply to give 63, then play with the signs. The roots are -7 and 9. The vast majority of equations do not have integer roots, but this exercise does focus attention on what they are.

The easiest derivation of the quadratic formula is by "completing the square". Start with

$$x^2 + \frac{b}{a}x + \frac{c}{a} = 0$$

We will first simplify the middle term, the cofactor of x. Remember that $m = -b/2a$ so $-2m = b/a$ and then plug in to get

$$x^2 - 2mx + \frac{c}{a} = 0$$

The bright idea is to turn this into $(x - m)^2$ which we do by adding and subtracting m^2:

$$x^2 - 2mx + m^2 - m^2 + \frac{c}{a} = 0$$

$$(x - m)^2 - m^2 + \frac{c}{a} = 0$$

The rest is simple algebra, which I leave to you.

Introductory examples

A quadratic equation contains the square of a variable multiplied by some constant, which may be 1. Often we call the variables y and x, so we have something like

$$y = ax^2$$

It may be that $a = 1$, and then $y = x^2$, but a cannot be zero, or we don't have a quadratic.

Then in addition there may be an optional term containing the first power of x and also perhaps a final constant. The general form is

$$y = ax^2 + bx + c$$

where a, b and c are constants. b or c (or both) might be zero.

The two most common applications are maximization problems, and gravity.

maximizing area

We need to build a fence or a wall from a fixed amount of material. There are constraints: it must be in the shape of a rectangle, and we want the area enclosed to be a maximum.

You will usually be given some definite amount of material, like 100 feet of fencing, but it does no harm to re-scale our problem, as long as we understand how to undo the scaling once we have the answer.

Let the perimeter of the rectangle be 2, the half-perimeter is 1, and then if one dimension, say the short side of the rectangle is h, the width is $1 - h$, so the area is

$$A = hw = h(1 - h) = -h^2 + h$$

As will be obvious later, since $a < 0$, the graph of this equation forms a "cup" that opens down. The value of A at the vertex is a maximum.

Two fundamental facts about parabolas:

- the maximum (or minimum) always occurs at the vertex
- the x-coordinate of the vertex has the value $-b/2a$

So the simplest parabola $y = x^2$ has $b = 0$, because there is no term like bx, and the vertex is at $x = 0$, $y = 0$.

In our problem $a = -1$ and $b = 1$, the vertex is the maximum and it is at $h = 1/2$. Since $h + w = 1$, it follows that $h = w = 1/2$.

The meaning of this answer is that the square ($h = w$) has the maximum area for a rectangular shape with a fixed perimeter.

Actually, you may know that if the shape is not rectangular the maximum area is for a circle or disk, but that is not the problem we're solving!

gravity

A second application is for objects falling under the influence of gravity. Gravity is an acceleration, and that acceleration is constant, which means that if the speed or velocity after falling for one second is v, the velocity after two seconds is twice v and so on.

So the distance $d = vt$ covered from zero time up to 1 second is less than that covered between $t = 1$ and $t = 2$ seconds.

Let me just give you the equation. If we drop our rock from the Tower of Pisa and the tower stands a height h_0 above the ground, the distance h above the ground after t seconds is

$$h = -16t^2 + h_0$$

This makes sense. At $t = 0$, $h = h_0$ and then h will be smaller than h_0 by some amount $-16t^2$ when $t > 0$.

If the initial height h_0 is 144 feet, then a dropped rock will hit the ground at $h = 0$ which means that

$$0 = -16t^2 + 144$$

$$t = \sqrt{144/16} = \sqrt{9} = 3 \text{ sec}$$

We take only the positive square root, but to explain that will take more time than we want for this example right now. We'll come back to it.

There is a limit to the final, or terminal velocity. It comes not from gravity but from air resistance. Someone who jumps from an airplane or gets thrown out through a window of the Nakatomi Plaza has a terminal velocity of about 180 feet per second or 120 miles per hour.

maximizing profit

You are the producer of a Broadway show. The theater seats 120 people and you have priced the tickets at \$10 each. You always fill the theater.

It costs you \$200 to rent the theater and another \$200 to pay the cast, so the fixed costs F are 400 and your profit for each show is

$$P = nT - F$$

$$= 120 \cdot 10 - 400 = 800$$

\$800 per show. That seems pretty good.

Your cousin Vinnie has an idea. If you raise the price you'll make more money, but you'll also sell fewer tickets. Acme Consulting has done a study which says that for every \$1 increase in the ticket price you'll have two empty seats.

As a red-blooded American, *of course* you want the maximum profit. What should you do? There's clearly money to be made, if you raise the ticket price \$1, you'll sell 119 seats at \$11 for \$1309, giving a net profit of \$909.

Call the price increase x. You will sell $n - 2x$ seats at a ticket price of $T + x$ for a profit

$$P = (n - 2x)(T + x) - 400$$

$$= -2x^2 + (n - 2T)x + nT - 400$$

The x corresponding to the maximum of this quadratic is $-b/2a$.

$$x = -(n - 2T)/2(-2)$$

$$= (120 - 20)/4 = 25$$

You increase the ticket price from \$10 to \$35. The profit will be

$$\begin{aligned} P &= (n-50)(T+25) - 400 \\ &= 70 \cdot 35 - 400 = 2050 \end{aligned}$$

That's an increase of \$1250 per show.

odd numbers

The product of two consecutive positive odd numbers is 255. Find the numbers.

Odd numbers can be described as $2n+1$ for some n, since that is one more than the even number $2n$. So we have

$$\begin{aligned} (2n+1)(2n+3) &= 255 \\ 4n^2 + 8n + 3 &= 255 \\ 4n^2 + 8n - 252 &= 0 \end{aligned}$$

We are looking for zeros. So we can divide by 4:

$$n^2 + 2n - 63 = 0$$

Two numbers multiply to give -63 and add to give 2:

$$(n+9)(n-7) = 0$$

The roots are $-9, 7$. But we specified positive numbers which means $n = 7$. The numbers are $2 \cdot 7 + 1 = 15$ and 17, easily confirmed.

Alternatively we can write the numbers as $2n+1$ and $2n-1$ and then

$$\begin{aligned} (2n+1)(2n-1) &= 255 \\ 4n^2 - 1 &= 255 \\ n^2 &= 64 \\ n &= \pm 8 \end{aligned}$$

Again $n > 0$ so $n = 8$ and the numbers are $2n-1 = 15$ and $2n+1 = 17$.

gravity

Many problems for quadratics involve gravity because the distance an object falls under gravity goes like t^2. That was Galileo's famous discovery. He used an inclined ramp and a water clock.

The business about dropping objects of different weights off the Tower of Pisa was a popular activity during the century preceding Galileo. His contribution was a thought experiment that is even more interesting.

`https://en.wikipedia.org/wiki/Galileo%27s_Leaning_Tower_of_Pisa_experiment`

In the metric system, the acceleration due to gravity (near the earth's surface) is very close to 10 meters per second^2. So, if we drop a rock, the distance traveled goes like

$$d = \frac{1}{2} 10\ t^2 = 5\ t^2$$

After 1 second, 5 meters, after 2 seconds 20 meters, and so on. If we know the distance to the ground and want the time:

$$t = \sqrt{\frac{d}{5}}$$

Some problems involve throwing an object up in the air to begin with, then we need the general equation for movement in one dimension.

$$x = \frac{1}{2} a t^2 + v_0 t + x_0$$

Which letter to use for the variable is your decision and really makes no difference. We can use y for height, by analogy with our graphs on the xy-axes. We can use h, for height, or d for distance. Or we can just use x, as we might in outer space.

x is really $x(t)$, the x-position depends on, it is *a function of*, the time.

The distance depends on the initial velocity v_0 times the time, which makes sense, and on the initial position x_0.

To use this equation we must keep our directions straight. Gravity accelerates downward, normally initial heights will be positive, and initial velocity is usually positive but may be negative. And of course the velocity may start positive but then surely turns negative, because gravity always wins.

You might be wondering about the other term, so consider a situation where v_0 and x_0 are both zero. Then

$$x = \frac{1}{2}at^2$$

There's a bit of a mystery here, why do we have the factor of 1/2, given the $v = at$?

The v in this formula is the velocity at the end of the interval, after a has acted to boost the velocity from v_0 to v.

The correct relationship *really* is

$$x = \bar{v}t$$

where $\bar{v}$ is the average velocity

$$\bar{v} = \frac{1}{2}at$$

We explore this a bit more in the next section.

time-dependence

Distance equals velocity times time.

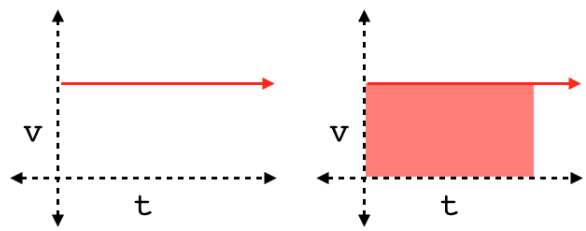
This is easy if the velocity is constant. Travel west on the interstate at exactly 60 miles per hour for 2 hours and your distance will be 120 miles from where you started (provided you don't start in Los Angeles).

It is standard to use s to refer to the distance traveled and v for velocity. If the velocity is constant then:

$$s = vt$$

According to the internet, s is from the Latin "spatium", for "space, room, or distance."

Suppose we plot velocity as a *function of time* with v on the y-axis and t on the x-axis.



Since the velocity is constant, the result is a straight horizontal line.

Furthermore, the distance traveled is the *area under the curve* (and above the x-axis) which is the area of a rectangle with sides v and t and as we said

$$s = vt$$

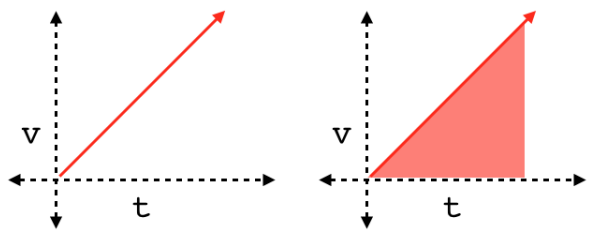
However, for most interesting problems the velocity is not constant.

Imagine maintaining pressure on the gas pedal in the car steadily so that, starting from a stop at zero time, after 1 second your velocity is 10 mph, after 2 seconds it is 20 mph, after 3 seconds, 30 mph. If we continue at the same rate of acceleration, we'll go from 0 to 60 mph in 6 seconds, which is quite a respectable time.

This example has constant acceleration. Here, we say that v is a constant function of time, and write

$$v = at$$

where a is the acceleration. It is the slope of the line.



What about the distance? It turns out that the distance is once again the area under the curve.

If a is non-zero and constant, then v changes at a constant rate. Starting from 0, the final velocity will be $v = at$, but the distance traveled is no longer the product

$$s = v \times t \stackrel{?}{=}$$

because this v is the final velocity and that is not the correct v to use. For variable velocity, the distance traveled is the *average* velocity times the time. For smooth (constant) acceleration from zero to v, the average velocity is the average of the initial and final velocities:

$$v_{\text{avg}} = \frac{1}{2}\,(v_i + v_f) = \frac{1}{2}\,v$$

So the correct equation is:

$$s = v_{\text{avg}}\,t = \frac{1}{2}\,v \cdot t$$

and since $v = at$

$$s = \frac{1}{2}at^2$$

In this case, if we plot velocity as a function of time, we obtain a straight line that extends diagonally up with respect to the x-axis. The distance traveled is the area under the curve, below the line and above the x-axis.

The shape whose area is needed is a triangle. This also accounts for the factor of 1/2.

$\bar{v}$ is the average velocity during the interval between 0 and t. Since the acceleration a is a constant, this is simply the average of the velocity at the end ($v = at$), and at the start ($v = 0$).

$$\bar{v} = \frac{1}{2}(v_f + v_0) = \frac{1}{2}(at + 0) = \frac{1}{2}at$$

So that's why there is a factor of one-half in the basic formula.

$$x = \frac{1}{2}at^2 + v_0 t + x_0$$

back to gravity

Again, the equation for gravity is

$$h = \frac{1}{2}gt^2 + v_0 t + h_0$$

This says that if we throw an object like a rock or a cannonball straight up into the air, the height at time t depends on the acceleration due to gravity g, the initial velocity v_0 in the upward direction, and the initial height h_0 relative to the earth's surface.

For gravity we have only two directions in one dimension, up and down. The constant g is negative while v_0 and h are positive, because the force of gravity points down while we have defined up as positive. We can rewrite this as

$$h = -16t^2 + vt + h_0$$

Here the units of velocity v are feet per second, h is in feet, and g is -32 feet per second per second. (If this were a physics book we'd use meters, and then g would be -10 meters per second per second.

Suppose the initial height is $h_0 = 96$, which means that you are almost 100 feet in the air, perhaps on a platform or the top of a cliff, and the initial velocity is upward at 16 feet per second. That's a little more than 20 miles per hour.

Clearly the rock will go up to some maximum height and then come down. The question asks first, at what time is the height a maximum? Recall the formula for the vertex.

$$m = -\frac{b}{2a} = -\frac{v_0}{2(-16)} = \frac{16}{32} = \frac{1}{2}$$

The rock reaches its maximum height at 1/2 second and that height is

$$h = -16(1/2)^2 + 16(1/2) + 96 = 112$$

My little brother throws 20 miles per hour. Suppose you could talk a professional baseball player into climbing up there with you and throwing vertically up at 64 feet per second. A fun fact is that 15 mph is exactly 22 feet per second.

$$64\,\frac{22}{15} \approx 95$$

about 95 miles per hour, a fastball.

Then

$$m = -\frac{v_0}{2(-16)} = \frac{-64}{-32} = 2$$

Two seconds to the maximum height.

If he throws the rock straight up, it will land on his head, or yours, in a stiff breeze. Suppose the player throws it outward a bit, but still up with the same velocity, so when it comes down it just misses his nose.

At what time will it reach the ground? We have

$$h = 0 = -16t^2 + 64t + 96$$

and we need the roots of this quadratic equation because we are asking for $h = 0$ which is ground level.

Do you remember where we said that the roots don't change if we factor out a constant?

$$h = 0 = (-16)(t^2 - 4t - 6)$$

We can't factor this one into integers. The times where $h = 0$ are, with $m = 2$ and $c = -6$:

$$m \pm \sqrt{m^2 - c} = 2 \pm \sqrt{4 + 6} = 2 \pm 3.16 = -1.16, 5.16$$

This says that the rock hits the ground after 5 seconds or so.

The negative root also can have a physical meaning under some conditions. The equation doesn't know about the baseball player or the ladder. All it knows is that we started our clock at zero with the rock at 96 feet above the ground and a vertical velocity of 16 feet per second upward. We won't worry about that.

If we are not allowed to use the simplification gained by factoring a and also told to use the quadratic equation, you will write

$$-16t^2 + 64t + 96 = 0$$

so

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-64 \pm \sqrt{64^2 - 4(-16)(96)}}{2(-16)}$$

I get 10240 for the discriminant and about 3.16 for the square root after dividing by 32. That matches the previous answer.

We can also say something about the velocity. The equation for velocity is

$$v = gt + v_0$$

The velocity at any time t is equal to the initial value plus the change due to gravity. Again, g is negative while v_0 is positive. Depending on time t, v may be either positive or negative. We had a factor of 1/2 before, that is $(1/2)g = -16$, but $g = -32$.

There is one moment when the velocity is zero, when the height is a maximum and the rock stops going up and comes back down.

In this problem

$$v = 0 = -32t + 64$$

Again, 2 seconds to maximum height.

At $t = 5.16$ the velocity is about

$$v = -32(5.16) + 64 = 100$$

maximizing distance

King Charles requires you to shoot a cannonball as far as possible. The distance depends on the angle at which the projectile is fired (we ignore air resistance).

This classic problem is usually solved by trigonometry, but we are going to do it directly. The velocity is some combination of horizontal and vertical components v_x and v_y

Using the Pythagorean theorem we have that

$$v^2 = v_x^2 + v_y^2$$

Aside from this relationship, the two component velocities act independently. If not for gravity the horizontal distance d would be

$$d = v_x t$$

and the clock would just keep ticking, so d would continue to increase, as it would if the cannon were fired in outer space.

However, the clock must stop, because for reasonable values of the velocity, gravity will bring the cannonball back down to earth before it leaves the atmosphere. That equation is

$$y = -16t^2 + v_y t + 0$$

We want the time when $y = 0$ so

$$16t^2 = v_y t$$

Since there are only two terms in this quadratic and they both contain t, we can simplify it by dividing out

$$16t = v_y$$

$$t = \frac{v_y}{16}$$

Dividing both sides by t in this way, we "lose" a solution, but that solution is $t = 0$, which we don't care about. Now plug that expression for t back into the first equation.

$$d = \frac{v_x v_y}{16}$$

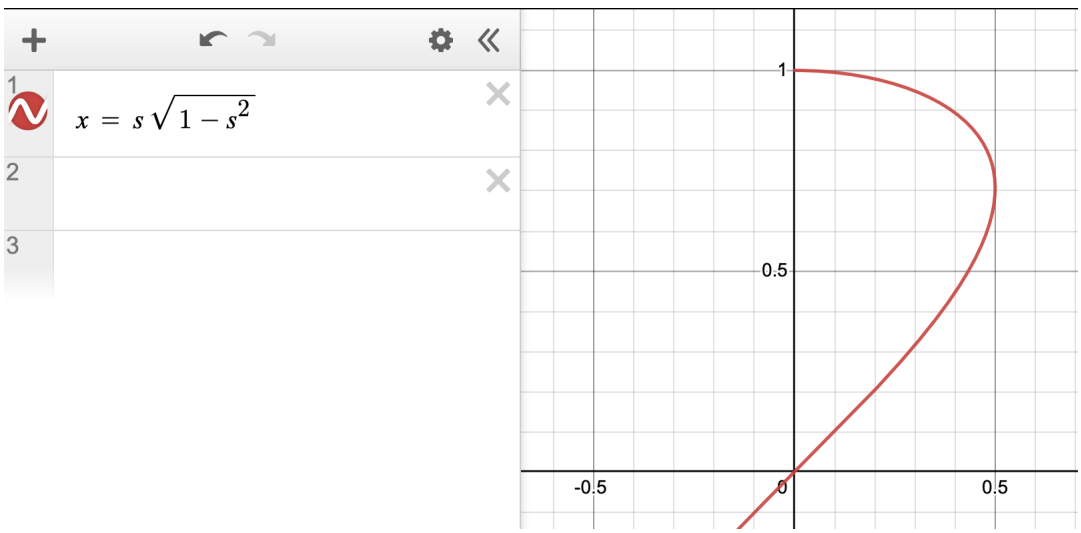
We need to choose v_x and v_y to maximize d. We can ignore the factor of $1/16$, since the value of v_x or v_y giving a maximum for d will also maximize $16d$.

In the end, we want to maximize

$$v_x v_y = v_x \sqrt{v^2 - v_x^2}$$

where v is a constant for a given cannon and amount of gunpowder, etc.

Just to explore a little, scale the problem so that $v = 1$, and come up with a simpler name for the variable v_x, like s. Then take a look in Desmos



The curve doesn't look like a standard parabola (it actually is one, but rotated by an angle). It seems that the maximum comes at roughly $s = 0.7$. That's a hint, because we know that $1/\sqrt{2} \approx 0.7$.

Without getting into the gory details, there is no reason to favor x over y. We expect they should be equal.

$$v^2 = v_x^2 + v_y^2 = 2v_x^2$$

$$v_x = \frac{v}{\sqrt{2}}$$

The maximum comes when $v_x = v_y$, that is, for an angle of 45°. I'm told this isn't true in real life, because of air resistance.

Putting this another way, to maximize this:

$$v_x\sqrt{v^2 - v_x^2}$$

we maximize the square:

$$v_x^2(v^2 - v_x^2)$$

v_x^2 will be some fraction f of v^2.

$$fv^2(v^2 - fv^2) = v^2\ [\ f(1-f)\ ]$$

v is a non-zero constant, so this problem really amounts to maximizing $f(1-f)$. The quadratic is

$$-f^2 + f = 0$$

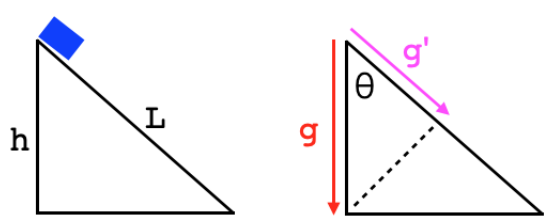
The maximum is $m = -b/2a = -(1)/(2)(-1) = 1/2$. Plug it into Desmos and take a look.

So v_x^2 is one-half of v^2 and equal to v_y^2. Hence $v_x = v_y$.

inclined plane

Think about a ramp or inclined plane at some angle θ to the vertical. Place a block of lead on the ramp and it slides down to the bottom. We ignore friction.

The vertical height is h, and the length of the ramp is L. It always helps to draw a picture.



It is a remarkable fact that forces (and most everything) can be converted easily between different coordinate systems.

The force along the ramp, making the block move, is the component of gravity acting in that direction. Using similar right triangles, it is easy to show that

$$\frac{h}{L} = \frac{g'}{g}$$

This is called the cosine of the angle θ but we don't need trigonometry for this problem.

The point is that a part of the force of gravity still acts along an inclined plane, multiplied by a factor less than one and equal to h/L. So our equation of motion is

$$s = \frac{1}{2}g't^2 + v_0 t + s_0$$

where $g' = gh/L$.

When does the block reach the bottom? Then, $s = L$ and

$$L = \frac{1}{2}g't^2 = 16h/L \cdot t^2$$

$$t = \frac{L}{4\sqrt{h}}$$

The velocity at the bottom is

$$v = g't = gh/L \cdot \frac{L}{4\sqrt{h}} = 8\sqrt{h}$$

Perhaps surprisingly, the answer is independent of L.

Different ramp lengths give the same final velocity, though not the same time.

If you know simple physics we can do this problem in another way, namely by conservation of energy.

$$P.E. = K.E.$$

The potential energy at the top is equal to the kinetic energy at the bottom.

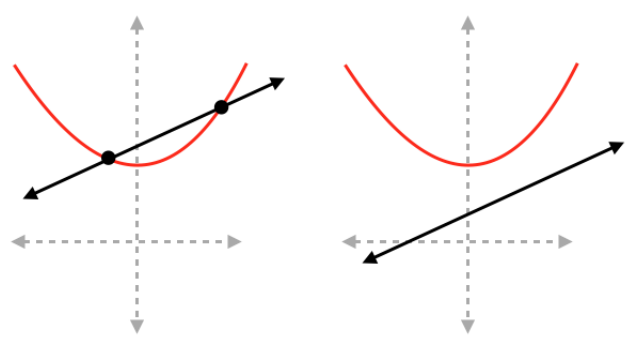
$$mgh = \frac{1}{2}mv^2$$
$$v^2 = 2gh = 64h$$
$$v = 8\sqrt{h}$$

which is what we had before.

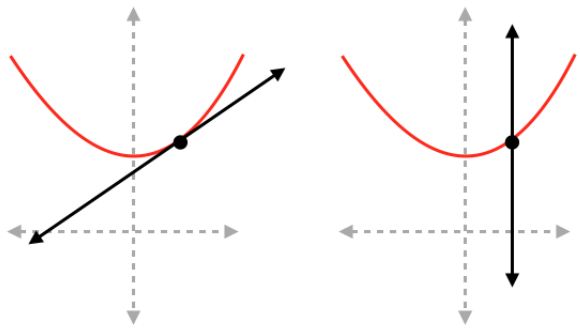
line and a parabola

Suppose we have generic parabola and a generic line in the plane, of just the sort we've been studying.

Here are the two most common possibilities. The line crosses the parabola at two points or it doesn't cross at all

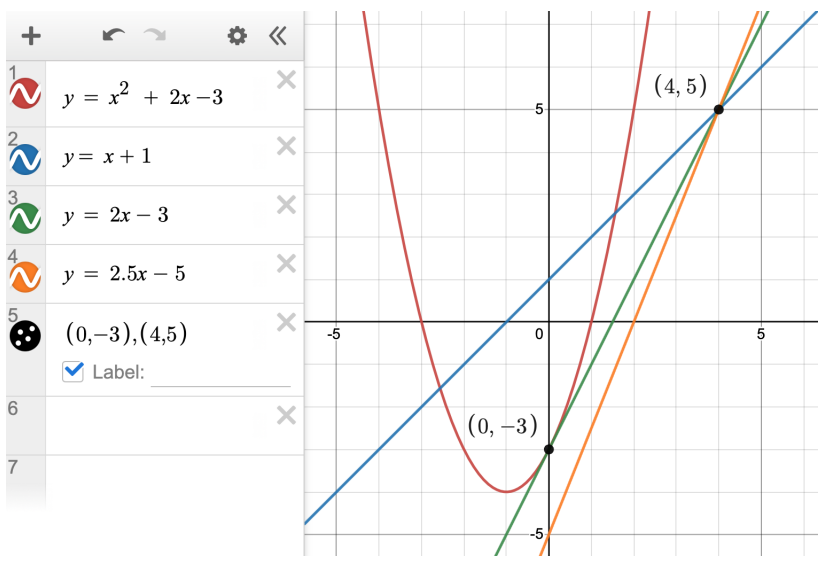


The other two possibilities both have a single point of intersection. The tangent line at a point (left panel), and a vertical line (right panel). In the latter case, the line has an undefined slope.



The line that just touches a curve at a point is called the tangent to the curve at that point, and this is a major topic of study in calculus.

Here's an example with actual numbers.



The parabola is in red with its vertex at $x = -b/2a = -1$ and roots at $x = -3, 1$. Three lines are shown, one crosses the curve at two points (blue), one does not cross at all (orange) and one is the tangent to the curve (green), touching only the point $(0, -3)$.

We will show how to find the equation of the green line easily.

As a reminder, suppose we have two lines

$$y = k_1 x + u$$

$$y = k_2 x + v$$

If we want to know where they meet, we are looking for some point (x, y) that solves both. Since the x's and the y's are equal we get

$$k_1 x + u = y = k_2 x + v$$

and we just solve for x.

The problem is to do the same thing with a quadratic and a line.

We have the line $y = kx + u$ and the parabola $y = ax^2 + bx + c$ and we set the two y's equal to obtain

$$kx + u = y = ax^2 + bx + c$$

Group terms

$$ax^2 + (b - k)x + (c - u) = 0$$

This is another quadratic and can be solved in the usual way, namely the roots are

$$x = m \pm \sqrt{m^2 - (c - u)/a)}$$

The crucial observation is this: the line that is *tangent* to the curve at (x, y) only touches at that single point. That's what tangent means.

Therefore, we are interested in the case where the equation we wrote has only one solution. That only happens if the discriminant is zero.

The square root is what gives us two solutions. If the square root goes away there is only one solution and we have therefore

$$x = m = -\frac{b - k}{2a}$$

$$2ax = -b + k$$
$$k = 2ax + b$$

This is the slope of the tangent line to a parabola at the point (x, y). The slope is proportional to x, which makes sense. The curve slopes more steeply upward, as x gets bigger.

I can't help pointing out that at the vertex, the slope is $k = 0$. So that means

$$2ax + b = 0$$
$$x = -\frac{b}{2a}$$

We have calculated m again (and got the same answer).

The tangent to the curve $ax^2 + bx + c$ has the slope $2ax + b$. You might guess that the 2 in the slope comes from the power of 2 in the parabola, and you'd be right.

note about i

We said that if $4ac > b^2$ we'll have the square root of a negative number. We called the term under the square root, D, the discriminant.

What people did was to invent a new kind of number i with the property that $i^2 = -1$, so $\sqrt{-1} = i$.

Then any negative square root can be written as $\sqrt{-N} = \sqrt{N} \cdot \sqrt{-1} = Ni$.

We had this as an example:

$$x^2 + 2x + 2 = 0$$

$a = 1$, $b = 2$ and $c = 2$. Then

$$b^2 - 4ac = 4 - 8 = -4$$

The roots are

$$x = -1 \pm \frac{\sqrt{-4}}{2} = -1 \pm i$$

If we write the equation out using the roots

$$(x - s)(x - t)$$

$$= (x - (-1 + i))(x - (-1 - i))$$

$$= ((x + 1) - i))((x + 1) + i)$$

we have four terms:

$$= (x + 1)^2 + i(x + 1) - i(x + 1) - i^2$$

The two terms in the middle cancel and the last one is +1:

$$= (x + 1)^2 + 1 = x^2 + 2x + 2$$

A real discussion about what i *means* will have to wait. But the point to notice is that when we have roots containing i they are always symmetric, it is always

$$(p + qi)(p - qi) = p^2 + q^2$$

And when multiplying out the terms with i cancel.

extra

You may want (or be required) to know the fancy version of the quadratic formula:

$$x = -\frac{b}{2a} \pm \frac{\sqrt{b^2 - 4ac}}{2a}$$

It is relatively easy convert between the two forms. Starting with

$$x = m \pm \sqrt{m^2 - c/a}$$

substitute for m

$$x = -\frac{b}{2a} \pm \sqrt{(\frac{b}{2a})^2 - c/a}$$

The key is now to put the last term $-c/a$ over the same denominator as the first term under the square root, $(2a)^2 = 4a^2$. To do that, multiply by $4a$ on top and bottom

$$x = -\frac{b}{2a} \pm \sqrt{(\frac{b}{2a})^2 - \frac{4ac}{4a^2}}$$

We have $4a^2$ in the denominator which can come out from under the square root and simplify to

$$x = -\frac{b}{2a} \pm \frac{\sqrt{b^2 - 4ac}}{2a}$$

However, I really prefer

$$x = m \pm \sqrt{m^2 - c/a}$$

especially since you must already know the formula for m, and the c/a part is visible in

$$x^2 + \frac{b}{a}x + \frac{c}{a} = 0$$